Alireza Modirshanechi

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Research Interests

My research lies at the intersection of **cognitive neuroscience** (*computational modeling*, *targeted experiment design*, and *model-based analysis of behavioral and neural data*) and **machine learning** (*reinforcement learning* and *Bayesian inference*). I formalize and organize psychological concepts—such as *surprise*, *curiosity*, and *sense of control*—to answer two fundamental questions: '**How do biological systems make intelligent decisions?**' and '**How can we design artificial systems that do the same?**'

Education

2018 – 2024 ■ **Ph.D. in Computer and Communication Sciences**
EPFL, Lausanne, Switzerland.
- *Thesis Committee*: W. Gerstner (Advisor), S.J. Gershman, J. Gold, and M. Weygandt Mathis.
- Featured on the EPFL news.

2014 – 2018 ■ **B.Sc. in Electrical Engineering (Communication)**
Sharif University of Technology, Tehran, Iran (GPA: 19.12/20).

Employment

2026 – ■ **Tenure-Track Professor**
University of Göttingen, Germany.
- Running the Laboratory of Natural and Artificial Intelligence (LNAI).

2024 – 2026 ■ **Postdoctoral Researcher**
Helmholtz Institute (Munich) and MPI for Biological Cybernetics (Tübingen), Germany.
- *Hosts*: Eric Schulz and Peter Dayan.
- Developed an integrative account of how natural (Modirshanechi et al., accepted at Nature Human Behaviour) and artificial systems (Modirshanechi et al., 2026, NeurIPS) sense and seek control and empowerment. Also co-supervised projects on developing human-like representations (Nath, Modirshanechi†, and Dayan†, 2026, PsyArXiv) and contributed to building foundation models of human cognition (Binz et al., 2025, Nature).

2018 – 2024 ■ **Ph.D. Student**
EPFL, Lausanne, Switzerland.
- *Supervisor*: Wulfram Gerstner.
- Developed a taxonomy of surprise and novelty definitions (Modirshanechi et al. 2022, Journal of Mathematical Psychology) and characterized their computational roles in human exploratory (Modirshanechi et al. 2025, PNAS) and adaptive behavior (Xu* and Modirshanechi* et al. 2021, PLOS Comp. Biology). Also worked on analyzing physiological and neuroimaging data (e.g., Esmaeili et al. 2021, Neuron).

Grants and Fellowships

2026 – 2032 ■ **The DFG Emmy Noether Programme**, ∼1.7M EUR.
Project: *The sense of control in natural and artificial intelligence*.

2018 ■ **EPFL EDIC Fellowship**, ∼51K CHF.

Selected Honors and Awards

2024 ■ Laureate of **the Dimitris N. Chorafas Foundation Award**.

2024 ■ Laureate of **the EPFL EDIC Thesis Distinction Award**.

2019 ■ Laureate of the annual **Teaching Assistant Award**, EPFL.

2018 ■ **Ranked 2nd** in the department's best B.Sc. thesis competition, Sharif University of Technology

2018 ■ **Ranked 2nd** in GPA among all entrants of 2014 (∼200 students), Sharif University of Technology

2014 ■ **Silver Medalist**, 45th International Physics Olympiad (IPHO), Kazakhstan.

2013 ■ **Gold Medalist**, 26th National Physics Olympiad, Iran.

Publications

∗: Joint first authors; †: Joint last authors; Underlined: Trainees working under my mentorship
Google Scholar profile: https://scholar.google.com/citations?user=ysDOYB8AAAAJ&hl=en

[1] A compositional model of semantic fluency
SS Nath, A **Modirshanechi**†, P Dayan†. (2026)
PsyArXiv preprint.

[2] Meta-learning as a principle for human-like visual representations
C Demircan, M Binz, A **Modirshanechi**, E Schulz. (2026)
arXiv preprint.

[3] Post-training makes large language models less human-like
M Binz, E Akata, A Almaatouq, ..., A **Modirshanechi**, ..., X Zhang, J Zhu, E Schulz. (2026)
arXiv preprint.

[4] An integrative framework for the human sense of control
A **Modirshanechi**, P Dayan†, E Schulz†. (2026)
Nature Human Behaviour (accepted)

[5] Unifying goal-conditioned RL and unsupervised skill learning via control-maximization
A **Modirshanechi**, B Eysenbach, P Dayan†, E Schulz†. (2026)
Proc. of the 40th Conf. on Advances in Neural Information Processing Systems (NeurIPS)
→ ***Oral presentation*** *(top 0.3% of submissions)*

[6] Representational similarity modulates neural and behavioral signatures of novelty
S Becker, A **Modirshanechi**, W Gerstner. (2026)
Neuron 114 (10): 1863-1876.e17.
→ *Preview by C Miehl and J Gjorgjieva: 'Novelty beyond counting'*

[7] Remembering the "When": Hebbian memory models for the time of past events
J Brea, A **Modirshanechi**, G. Iatropoulos, W Gerstner. (2026)
PLOS Computational Biology 22 (7): e1014575.

[8] Learning is a fundamental source of behavioral individuality
R Manna, J Brea, GV Braga, A **Modirshanechi**, I Tomic, AM Jaksic. (2026)
eLife 15, RP111235.

[9] Novelty as a drive of human exploration in complex stochastic environments
A **Modirshanechi**, W Lin, HA Xu, MH Herzog, W Gerstner. (2025)
Proceedings of the National Academy of Sciences (PNAS) 122 (39): e2502193122.

[10] A foundation model to predict and capture human cognition
M Binz, E Akata, M Bethge, ..., A **Modirshanechi**, ..., DU Wulff, H Xiong, E Schulz. (2025)
Nature, 644: 1002–1009.

[11] Merits of curiosity: a simulation study
L Gruaz∗, A **Modirshanechi**∗, S Becker, J Brea. (2025)
Open Mind 9: 1037–1065.

[12] Computational models of intrinsic motivation for curiosity and creativity
S Becker, A **Modirshanechi**, W Gerstner. (2024)
Behavioral and Brain Sciences, 47: e94.

[13] Distributed and specific encoding of sensory, motor, and decision information in the mouse neocortex during goal-directed behavior
A Oryshchuk, C Sourmpis, ..., A **Modirshanechi**, W Gerstner, CCH Petersen†, S Crochet†. (2024)
Cell Report, 43: 113618.

[14] Curiosity-driven exploration: foundations in neuroscience and computational modeling
A **Modirshanechi**, K Kondrakiewicz, W Gerstner, S Haesler. (2023)
Trends in Neurosciences 46 (12): 1054–1066.

[15] Surprise and novelty in the brain
A **Modirshanechi**, S Becker, J Brea, W Gerstner. (2023)
Current Opinion in Neurobiology 82: 102758.

[16] A taxonomy of surprise definitions
A **Modirshanechi**, J Brea, W Gerstner. (2022)
Journal of Mathematical Psychology 110: 102712.

[17] Brain signals of a Surprise-Actor-Critic model: evidence for multiple learning modules in human decision making
V Liakoni∗, MP Lehmann∗, A **Modirshanechi**, J Brea, A Lutti, W Gerstner†, K Preuschoff†. (2022)
NeuroImage, 246: 118780.

[18] Novelty is not Surprise: human exploratory and adaptive behavior in sequential decision-making
HA Xu*, A **Modirshanechi***, MP Lehmann, W Gerstner†, MH Herzog†. (2021)
PLOS Computational Biology 17 (6): e1009070.

[19] Learning in volatile environments with the Bayes Factor surprise
V Liakoni*, A **Modirshanechi***, W Gerstner, J Brea. (2021)
Neural Computation 33 (2): 269–340.

[20] Fitting summary statistics of neural data with a differentiable spiking network simulator
G Bellec*, S Wang*, A **Modirshanechi**, J Brea†, W Gerstner†. (2021)
Proc. of the 34th Conf. on Advances in Neural Information Processing Systems (NeurIPS): 18552–18563.

[21] Rapid suppression and sustained activation of distinct cortical regions for a delayed sensory-triggered motor response
V Esmaeili*, K Tamuras*, SP Muscinelli, A **Modirshanechi**, ..., W Gerstner, CCH Petersen. (2021)
Neuron, 109 (13): 2183–2201.e9.

[22] Trial-by-trial surprise-decoding model for visual and auditory binary oddball task
A **Modirshanechi**, MM Kiani, H Aghajan. (2019)
NeuroImage, 196: 302–317.

[23] Election vote share prediction using a sentiment-based fusion of Twitter data with Google Trends and online polls
P Kassraie*, A **Modirshanechi***, H Aghajan. (2017)
Proc. of the 6th International Conf. on Data Science, Tech. and Applications (DATA 2017): 363–370.

Conference Talks and Posters

Contributed Talks (Oral Presentation):

[1] Unifying goal-conditioned RL and unsupervised skill learning via control-maximization
A **Modirshanechi**, B Eysenbach P Dayan†, E Schulz†
NeurIPS 2026 (satellite event in Paris)

[2] Seeking a sense of control: A normative perspective
A **Modirshanechi**, P Dayan†, E Schulz†
'Principles of Intrinsic Motivations in Biological and Artificial Agents' workshop at RLDM 2025 (Dublin)

[3] Representational similarity modulates neural and behavioral signatures of novelty
S Becker, A **Modirshanechi**, W Gerstner
Cosyne 2024 (Lisbon)

[4] Novelty drives exploration, whereas surprise modulates learning
A **Modirshanechi**, HA Xu, MP Lehmann, MH Herzog, W Gerstner
Bernstein 2021 (Online)

Selected Posters:

[1] An integrative framework for the sense of control
A **Modirshanechi**, P Dayan†, E Schulz†
CCN 2025 (Amsterdam), Cosyne 2026 (Lisbon), SBDM 2026 (Paris), and NeuroMonster 2026 (Rome)

[2] A principled framework for individual differences in neural networks
M Mathony, A **Modirshanechi**†, E Schulz†
NeuroMonster 2026 (Rome) and CogSci 2026 (Rio)

[3] A compositional model of semantic fluency
SS Nath, A **Modirshanechi**†, P Dayan†
CCN 2026 (New York)

[4] Seeking a sense of control: a normative perspective
A **Modirshanechi**, P Dayan†, E Schulz†
RLDM 2025 (Dublin)

[5] Merits of curiosity: a simulation study
L Gruaz*, A **Modirshanechi***, S Becker, and J Brea
RLDM 2025 (Dublin)

[6] Novelty drives human exploration even when it is suboptimal
A **Modirshanechi**, HA Xu, W Lin, MH Herzog, W Gerstner
Cosyne 2023 (Montreal), CCN 2023 (Oxford), & Bernstein 2023 (Berlin)

[7] Environmental similarity structure modulates neural and behavioral signatures of novelty
S Becker, A **Modirshanechi**, W Gerstner
Bernstein 2023 (Berlin) and RLDM 2025 (Dublin)

[8] Theories of surprise: definitions and predictions
A **Modirshanechi**, J Brea, W Gerstner
Cosyne 2022 (Lisbon)

[9] Model-based surprise modulates model-free learning
A **Modirshanechi**[*], HA Xu[*], MP Lehmann, MH Herzog[†], W Gerstner[†]
Cosyne 2021 (Online)

Selected Invited Talks

2027 ■ Bernstein Center for Computational Neuroscience (BCCN), Berlin, Germany.
2026 ■ Munich Interactive Intelligence Initiative (MI3), LMU, Munich, Germany.
2026 ■ The symposium on Modeling the Predictive Brain, PuG conference, Heidelberg, Germany.
2026 ■ The Simons Collaboration on Ecological Neuroscience (SCENE; Online).
2026 ■ Joint meeting of Chris Donkin and Thorsten Pachur Labs, LMU-TUM, Munich, Germany.
2026 ■ Faculty of Mathematics and Computer Science, University of Göttingen, Germany.
2025 ■ Curiosity, Information Seeking, & Exploration Conference, Brown University, USA (Online).
2025 ■ Chris Summerfield Lab, Oxford, UK (Online).
2024 ■ Cognitive Neuro. Colloquium, Uni. Medical Center Hamburg-Eppendorf, Germany.
2024 ■ Charley Wu Lab, University of Tübingen, Germany (Online).
2024 ■ Sony CSL Paris, France (Online).
2024 ■ Matthias Gruber Lab, Cardiff University, UK (Online).
2024 ■ Theory kick-off of RTG Curiosity program in the University of Göttingen, Germany.
2024 ■ Institut de Neurosciences de la Timone (INT), Marseille, France.
2023 ■ Computational Phenomenology Discussion Group (Online).
2023 ■ Angelika Steger Lab, ETH Zurich, Switzerland.
2023 ■ Active Inference Institute (Online).
2022 ■ Eric Schulz Lab, MPI for Biological Cybernetics, Tübingen, Germany.
2022 ■ Sebastian Haesler Lab, Katholieke Universiteit Leuven, Belgium (Online).
2022 ■ Ilya Monosov Lab, University of Washington in St. Louis, US (Online).
2022 ■ Marcelo Mattar Lab, UC San Diego, US (Online).
2022 ■ Montreal AI-Neuro Seminar Series, Montreal, Canada (Online).
2021 ■ The Allen Institute, Seattle, US (Online).
2021 ■ EPFL Neuro Symposium, Lausanne, Switzerland (Online).

Teaching Experience

2026 – ■ **Professor (Course Instructor)**
University of Göttingen, Germany
- Seminar on "Multi-Agent Reinforcement Learning"

2019 – 2023 ■ **Teaching Assistant**
EPFL, Lausanne, Switzerland.
- Artificial Neural Networks and Reinforcement Learning ($\times 4$).
  $\rightarrow$ Taught a series of lectures on "Curiosity-Driven Exploration."
- Markov Chains and Algorithmic Applications ($\times 3$).
  $\rightarrow$ Received the Department Teaching Assistant Award.
- Probability and Statistics.

2015 – 2018 ■ **Teaching Assistant**
Sharif University of Technology, Tehran, Iran.
- Computational Neuroscience.
- Communication Systems.
- Engineering Mathematics ($\times 2$).
- Signals and Systems.
- Electromagnetism.

2013 – 2017 ■ **Physics Olympiad Instructor**
National Organization for the Development of Exceptional Talents, Iran.
- Preparing high-school students for national and international Physics Olympiads.

Mentorship and Supervision

Ph.D. Students

2025 – ■ Nan Chen
MPI for Biological Cybernetics; co-supervised with Peter Dayan

2024 – ■ Surabhi S Nath
MPI for Biological Cybernetics; co-supervised with Peter Dayan

2024 – ■ Marvin Mathony
Helmholtz Munich; co-supervised with Eric Schulz

2023 – 2024 ■ Lucas Gruaz
EPFL; co-supervised with Wulfram Gerstner and Johanni Brea

2021 – 2024 ■ Sophia Becker
EPFL; co-supervised/mentored with Wulfram Gerstner

M.Sc. Students

2022 ■ Yehya El Hassan
EPFL; co-supervised with Johanni Brea

2022 ■ Auguste Poiroux
EPFL; co-supervised with Johanni Brea

2020 ■ Anthony Jakob
EPFL

2020 ■ Lucas Gruaz
EPFL

2019 ■ Alexandru Mocanu
EPFL; co-supervised with Martin Barry

2018 ■ Manu Srinath Halvagal
EPFL; co-supervised with Martin Barry

B.Sc. Students

2017 – 2018 ■ Mahdiyar Shahbazi
Sharif University of Technology

2017 – 2018 ■ Alireza Shirzad
Sharif University of Technology

2017 – 2018 ■ Amir Hoseini
Sharif University of Technology

2017 – 2018 ■ Ali Shirali
Sharif University of Technology

2016 – 2018 ■ Erfan Khalaj
Sharif University of Technology

Service and Volunteer Work

2020 – ■ Author of **educational articles** in Medium Publications; selected articles:
- 'Why does the optimal policy exist?' (more than 15K views)
- 'Is correlation distance a metric?' (more than 10K views)
- 'What can we learn from posterior distributions?' (more than 2K views)

2024 – ■ **Adhoc reviewer**:
- Journals (alphabetically ordered):
  - Biological Psychiatry: CNNI
  - Cognition
  - Computational Brain & Behavior
  - Current Opinion in Behavioral Sciences
  - Experimental Brain Research
  - Frontiers in Psychology
  - Journal of Mathematical Psychology
  - Learning and Motivation
  - Minds and Machines
  - Nature Communications
  - PLOS Computational Biology
  - PNAS
  - PNAS Nexus
  - Psychological Review
  - Scientific Reports
  - Trends Open

- Conferences and workshops (alphabetically ordered):
  - CCN 2023 & 2025
  - Cosyne 2026
  - NeurIPS Workshop: Data on the Brain & Mind 2025
  - NeurIPS 2026 (*Recognized as 'Top Reviewer'*)

2025 ■ **Workshop co-organizer** at CCN 2025 (Amsterdam, The Netherlands)
Title: 'What benchmarks are useful for cognitive science?'

2024 ■ **Workshop co-organizer** at Cosyne 2024 (Lisbon, Portugal)
Title: 'The predictive brain: from algorithms to mechanisms'

2022 ■ **Workshop co-organizer** at Bernstein 2022 (Berlin, Germany)
Title: 'Surprise in the brain: Theory and Experiments'

2022 – 2024 ■ **Volunteer** in Review of Application Materials (RAMP) program at EPFL to help prospective Ph.D. students with their applications (e.g., reviewing CVs, cover letters, etc.).

2020 – 2024 ■ **Volunteer** in the EPFL Buddy program (every year) to help 1st-year Ph.D. students at EPFL with starting their Ph.D (e.g., EPFL courses, life in Lausanne, etc.).